

ZOKA

Privacy by Default Blockchain

VERSION 1.0

Abstract

ZOKA is a privacy-first blockchain architecture that combines:

1. Groth16 zk-SNARK proofs over BLS12-381.
2. Pedersen commitments for confidential values.
3. Bulletproof range proofs.
4. CLSAG ring signatures over Ristretto255.
5. Stealth addresses with scan and spend key separation.
6. PBFT consensus with deterministic finality.
7. Dandelion++ plus Tor or I2P transport privacy.

This document specifies the technical model, security assumptions, anonymity functions, and measurable privacy metrics. The scope is engineering-grade protocol documentation. No marketing claims are included.

1. Scope and Status

This whitepaper describes Version 1.0 protocol behavior and implementation direction.

- External cryptographic audit is required before public mainnet.
- MPC ceremony workflow is required before public mainnet.
- No claim is made here that these milestones are already completed.

2. Threat Model

The system is designed against:

1. Passive chain observer.
2. Malicious validator (up to PBFT tolerance).
3. Passive network observer.
4. Node operator misconfiguration risk.
5. Local disk compromise without wallet password.

Out of scope:

- Post-quantum adversaries.
- Perfect anonymity against a global, real-time, full-network observer.

3. Cryptographic Stack

3.1 Curves and Domains

- BLS12-381: SNARK proving and verification.
- Ristretto255: ring signatures, commitments, stealth keys, range proof commitments.

3.2 Pedersen Commitments

For value v and blind r :

$$C(v, r) = v \cdot G + r \cdot H$$

with independent generators G, H .

Properties:

- Perfect hiding.
- Computational binding.
- Additive homomorphism.

3.3 Confidential Balance Equation

For each private transaction:

$$\text{sum}(C_{\text{in}}) = \text{sum}(C_{\text{out}}) + C_{\text{fee}} + X \cdot H$$

where:

- $C_{\text{fee}} = \text{fee} \cdot G + r_{\text{fee}} \cdot H$
- $X = \text{sum}(r_{\text{in}}) - \text{sum}(r_{\text{out}}) - r_{\text{fee}}$

Validation fails if conservation does not hold.

3.4 Range Proofs

Each committed amount is constrained to:

$$0 \leq v < 2^{32}$$

via Bulletproof verification.

3.5 Nullifiers

Each consumed note yields a nullifier:

$$N_i = H(s_i \parallel idx_i \parallel nonce_i)$$

Double spend prevention is enforced by nullifier-set uniqueness.

3.6 CLSAG Ring Signatures

For ring size N , signature payload layout is:

$$\text{sig} = c_0 \parallel \text{key_image} \parallel s_0 \parallel s_1 \parallel \dots \parallel s_{(N-1)}$$

Serialized size:

$$L_{\text{sig}}(N) = 32 \cdot (N + 2) \text{ bytes}$$

Verification is linear in N .

4. Transaction and Validation Pipeline

Private transaction carries:

- ring metadata (**ring_members**, **ring_signature**, **ring_version**)
- input nullifiers and input commitments
- output commitments and range proofs
- fee commitment and fee range proof
- SNARK proof and root anchors

Validation sequence:

1. Structural checks.
2. Nullifier uniqueness.
3. Ring signature verification.
4. Range proof checks.
5. Commitment balance equation.
6. Inclusion proof checks.
7. SNARK verification.

Active runtime path

ring_version = 1 is the active runtime path for private ring validation.

5. Consensus and Transport

5.1 PBFT

PBFT tolerates:

$$f = \text{floor}((n - 1)/3)$$

Byzantine validators in a fixed validator set.

5.2 Dandelion++ and Anonymous Transport

- Stem phase: single-path relay.
- Fluff phase: gossip broadcast.
- Tor/I2P support for transport-layer anonymity.
- Policy can enforce anonymous transport to block direct leakage.

5.3 Mixer Batching

Multiple private transactions can be emitted in one batch to reduce timing linkability.

6. Anonymity Functions and Metrics

This section defines measurable privacy functions.

6.1 Sender Anonymity Function

Let:

- **N** = ring size.
- **q** in **(0, 1]** = decoy quality factor (1 means ideal decoys).

Effective ring size:

$$N_{\text{eff}} = 1 + q \cdot (N - 1)$$

Best posterior guess:

$$P_{\text{sender}} = 1 / N_{\text{eff}}$$

Sender entropy in bits:

$$H_{\text{sender}} = \log_2(N_{\text{eff}})$$

6.2 Network Source Exposure Function

Let:

- **c** = fraction of adversarial first-hop candidates.
- **h** = stem hop count.

Approximate stem compromise probability:

$$P_{\text{stem_comp}}(c, h) = 1 - (1 - c)^h$$

Network anonymity score:

$$A_{\text{net}} = 1 - P_{\text{stem_comp}}$$

6.3 Timing Linkability Function

Let:

- **b** = effective batch cardinality seen by observer.

If observer cannot order sender inside batch:

$$P_{\text{time_link}} = 1 / b$$

$$H_{\text{time}} = \log_2(b)$$

6.4 Composite Privacy Index

Define normalized components in **[0, 1]**:

$$S = H_{\text{sender}} / H_{\text{ref}}$$

$$T = H_{\text{time}} / H_{\text{ref}}$$

$$N_w = A_{\text{net}}$$

with reference entropy $H_{\text{ref}} = \log_2(64) = 6$.

Composite index:

$$PI = w_s \cdot S + w_n \cdot N_w + w_t \cdot T$$

Example weights:

- $w_s = 0.45$
- $w_n = 0.35$
- $w_t = 0.20$

7. Numerical Examples

7.1 Example A (Current profile)

Assume:

- $N = 11$
- $q = 1.0$
- $h = 2$
- $c = 0.15$
- $b = 16$

Calculations:

$$\begin{aligned}
 N_{\text{eff}} &= 11 \\
 P_{\text{sender}} &= 1/11 &= 0.0909 \\
 H_{\text{sender}} &= \log_2(11) &= 3.459 \\
 P_{\text{stem_comp}} &= 1 - 0.85^2 &= 0.2775 \\
 A_{\text{net}} &= 0.7225 \\
 P_{\text{time_link}} &= 1/16 &= 0.0625 \\
 H_{\text{time}} &= 4.000
 \end{aligned}$$

Normalized:

$$\begin{aligned}
 S &= 3.459/6 = 0.5765 \\
 T &= 4/6 = 0.6667
 \end{aligned}$$

Composite:

$$PI = 0.45 \cdot 0.5765 + 0.35 \cdot 0.7225 + 0.20 \cdot 0.6667 = 0.6456$$

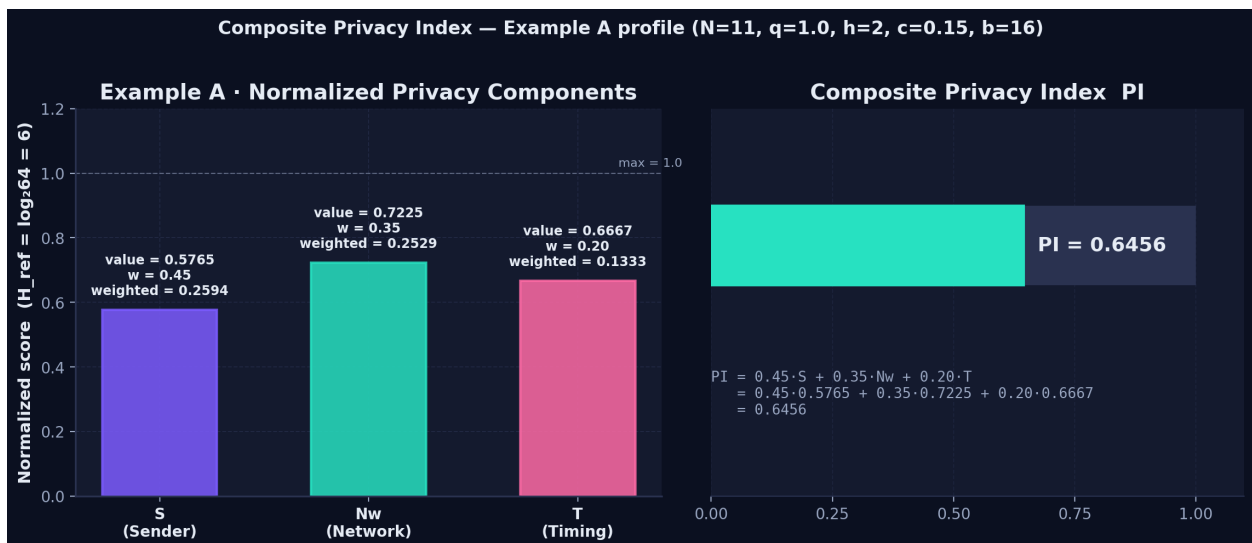


Figure · Example A — Normalized components and composite privacy index.

7.2 Example B (Degraded decoy quality)

Assume $N = 11$, $q = 0.7$.

$$\begin{aligned} N_{\text{eff}} &= 1 + 0.7 \cdot 10 = 8 \\ P_{\text{sender}} &= 1/8 = 0.125 \\ H_{\text{sender}} &= \log_2(8) = 3.000 \end{aligned}$$

Sender privacy declines even with same nominal ring size.

8. Graphs of Relevant Anonymity Functions

Graphs are provided as deterministic sampled points for direct plotting in Python, GNUplot, R, or spreadsheet tools.

8.1 Graph G1: Sender Entropy vs Ring Size ($q = 1.0$)

Function: $H_{\text{sender}}(N) = \log_2(N)$

N	H_sender (bits)	P_sender
2	1.000	0.5000
4	2.000	0.2500
8	3.000	0.1250
11	3.459	0.0909
16	4.000	0.0625
24	4.585	0.0417
32	5.000	0.0313

ASCII trend:

N: 2 4 8 11 16 24 32
H: * - - - * - - - * - - - * - - - * - - - * - - - * - - - *

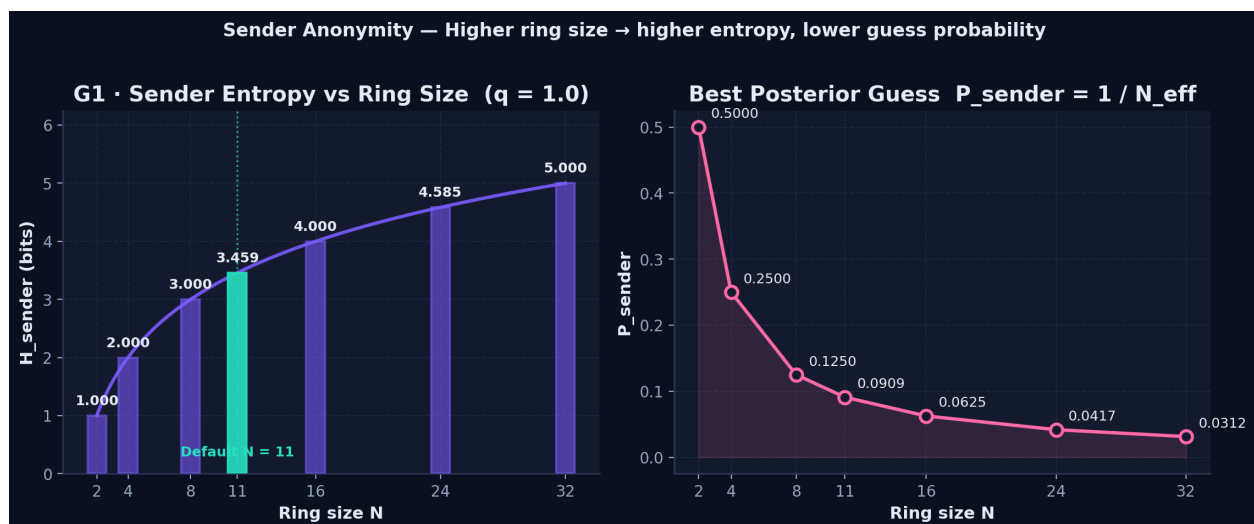


Figure G1 · Sender entropy grows with ring size; posterior guess falls.

8.2 Graph G2: Stem Compromise Probability vs Adversarial Fraction

Function with $h = 2$:

$$P_{\text{stem_comp}}(c) = 1 - (1 - c)^2$$

c	P_stem_comp	A_net
0.05	0.0975	0.9025
0.10	0.1900	0.8100
0.15	0.2775	0.7225
0.20	0.3600	0.6400
0.30	0.5100	0.4900
0.40	0.6400	0.3600
0.50	0.7500	0.2500

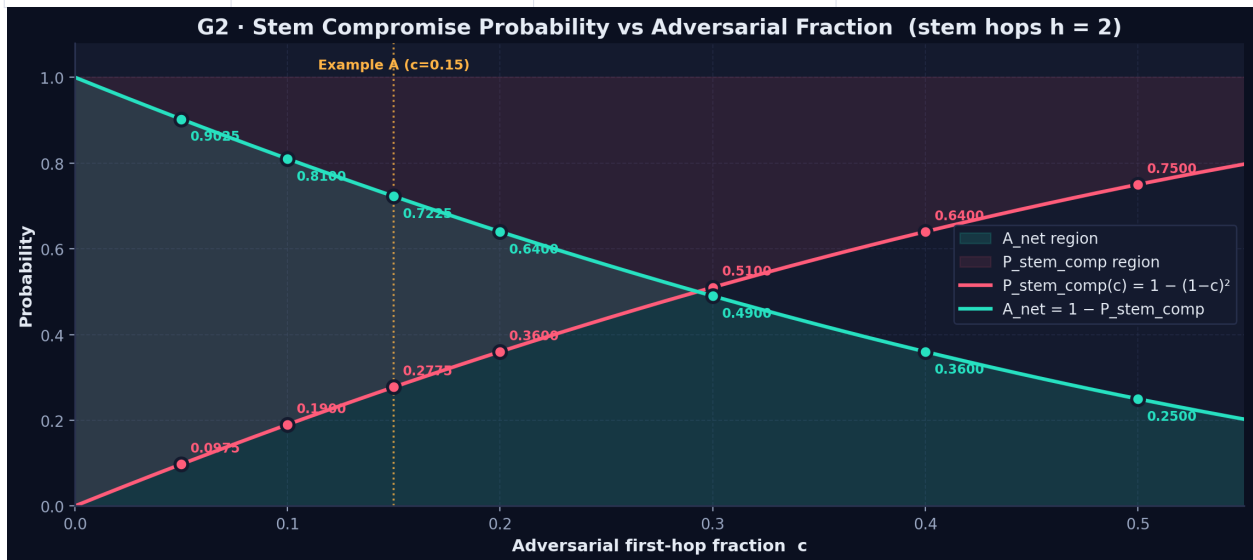


Figure G2 · As the adversarial fraction rises, stem compromise grows and network anonymity shrinks symmetrically.

8.3 Graph G3: Timing Linkability vs Batch Size

Function:

$$P_time_link(b) = 1/b$$

$$H_time(b) = \log_2(b)$$

b	P_time_link	H_time (bits)
1	1.0000	0.000
2	0.5000	1.000
4	0.2500	2.000
8	0.1250	3.000
16	0.0625	4.000
32	0.0313	5.000

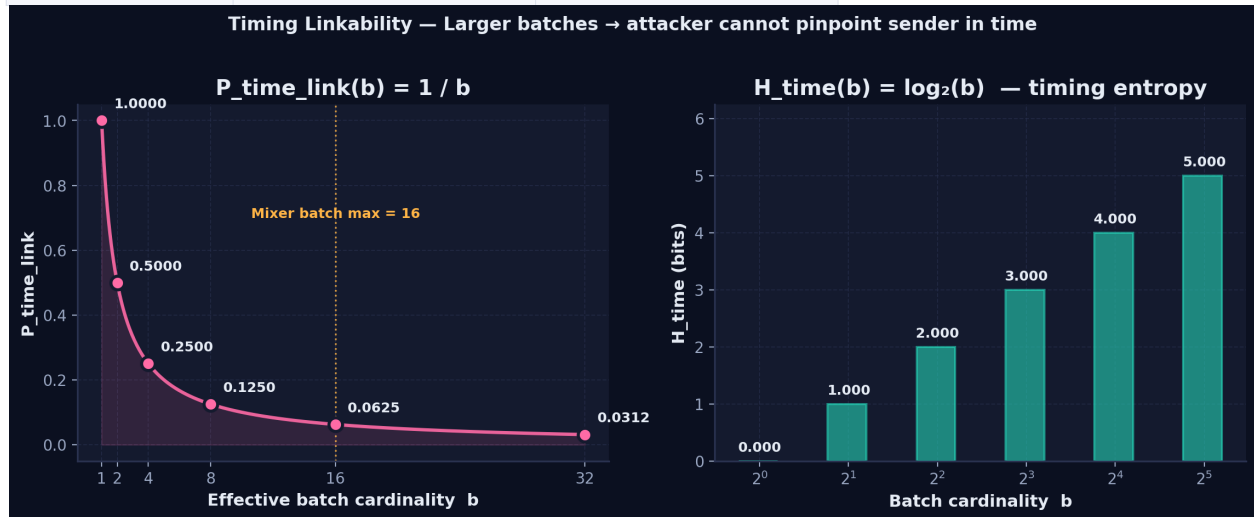


Figure G3 · Larger batches reduce timing linkability and add timing entropy.

9. Engineering Parameters (Version 1.0)

Parameter	Value
SNARK curve	BLS12-381
Ring curve	Ristretto255
Private tree depth	16
Max inputs per private tx	8
Outputs per private tx	2
Default ring size target	11
Bulletproof amount range	32-bit
Default stem hops	2
Mixer batch max	16
Mixer hold time	2000 ms

10. Limitations and Required Pre-Mainnet Work

Mandatory before public mainnet:

1. MPC ceremony completion for production CRS.
2. Independent external cryptographic audit.
3. Protocol freeze and reproducible release process.

11. Implementation Mapping

Current repository binaries:

- **zoka**
- **user-node**
- **p2p_worker**
- **generate_trusted_setup**
- **user-cli**
- **user-cli-private**

Planned but not currently shipped in Version 1.0:

- **zoka-ceremony-contribute**

12. Conclusion

ZOKA combines confidential transaction algebra, ring-based sender ambiguity, stealth receiver privacy, and transport obfuscation in one protocol. The privacy model is quantifiable through explicit

functions (**H_sender**, **P_stem_comp**, **P_time_link**) and a composite metric (**PI**) suitable for regression tracking across releases.

This document defines Version 1.0 as a technical baseline.

References

- [1] Groth, J. (2016). On the Size of Pairing-Based Non-interactive Arguments.
- [2] Bunz et al. (2018). Bulletproofs: Short Proofs for Confidential Transactions and More.
- [3] Goodell, Noether, RandomRun (2019). Concise Linkable Ring Signatures.
- [4] Castro, Liskov (1999). Practical Byzantine Fault Tolerance.
- [5] Fanti et al. (2018). Dandelion++.
- [6] de Valence, Yun, Grigg (2019). The Ristretto Group.